

Manual Lathe Machining Analysis

Cutting parameters, cutting force, and heat generation in turning and drilling

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Course	ME 215-102, NJIT	Instructor	Prof. Suwattananont
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PROJECT SUMMARY

This project introduced the fundamentals of conventional machining using a manual lathe and milling machine. After a review of shop safety, the instructor demonstrated turning, facing, drilling, shaping, and broaching operations using manual controls rather than CNC programming. The session emphasized accurate positioning along the x, y, and z axes and showed how adjusting feed rate, depth of cut, and spindle speed changes the result. A force sensor was calibrated and used to record cutting loads for four cuts at a constant 1,100 RPM, and a temperature rise test tracked heat buildup in the drill and workpiece. Cutting force increased with both feed and depth of cut, and friction was confirmed as the main source of heat during cutting.

1. Introduction & Objectives

This project covered the operation of a lathe and a manual milling machine, and showed how small changes to tool shape, cutting force, and speed significantly affect the appearance, feel, and function of a finished part. The objectives were to:

- understand the principles of machining and machining analysis;
- see how cutting power, speed, and other parameters influence part quality;
- learn the techniques needed to operate the equipment; and
- apply the safety practices required around rotating machinery.

2. Background

Machining removes material from metals and other stock to produce a desired shape. A lathe rotates the workpiece while a stationary tool cuts it away. Several lathe tools were used during the session: single-point cutting tools for turning, multi-point tools with several cutting edges, and form tools that imprint a specific profile onto a metal cylinder. Tool geometry is critical — the rake, clearance, and wedge angles must be correct before a part can be machined properly, because they directly control cutting force and how the metal deforms into a chip.

The three primary lathe operations were **turning**, where the tool travels along the rotating part and reduces its diameter; **facing**, where the tool travels across the end of the part; and **drilling**, where a tool held in the tailstock produces a hole on the axis of rotation.

3. Procedure

The session began with a safety briefing covering eye protection and safe distances around the machine, followed by an overview of machine components and terminology while the instructor ran the lathe. A force sensor was first calibrated by hanging known weights and recording the voltage output. Four cuts were then taken at fixed axial positions along the sample, alternating between two feed settings and two depths of cut at a constant spindle speed, while the sensor output was recorded. Finally, temperature was logged against time for both the drill and the top of the workpiece during a drilling pass.

4. Results

Table 1. Force sensor calibration data

Load (lb)	2	12	22	32	42
Output (mV)	-0.508	-0.444	-0.381	-0.317	-0.259

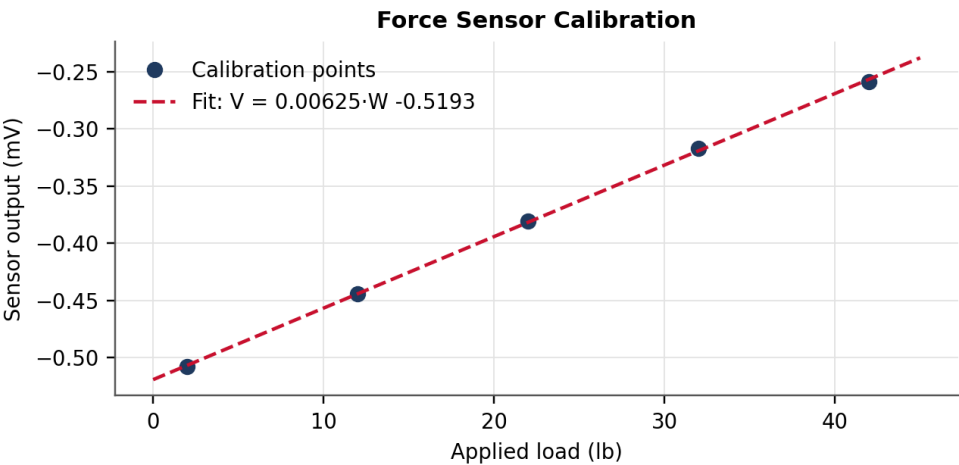


Figure 1. Force sensor calibration. The response is highly linear ($R^2 \approx 0.9997$), about 0.00625 mV per lb.

Using the calibration line, each cut’s sensor reading was converted into an approximate cutting force (Table 2).

Table 2. Machining parameters and estimated cutting force

Cut	Axial position z (in)	Depth, x (in)	Feed setting	Spindle (RPM)	Output (mV)	Est. force (lb)
1	2.016	0.015	40	1,100	-0.444	≈ 12
2	1.500	0.015	80	1,100	-0.275	≈ 39
3	0.999	0.007	40	1,100	-0.486	≈ 5
4	0.500	0.007	80	1,100	-0.403	≈ 19

Table 3. Temperature rise during drilling

Time (min)	0	0.268	0.618	0.851	1.254	1.452	1.678	1.852	2.128
Workpiece (°C)	24.1	27.4	31.7	43.5	36.3	33.9	36.1	39.8	41.7
Drill (°C)	24	32	40	48	49	52	55	58	66

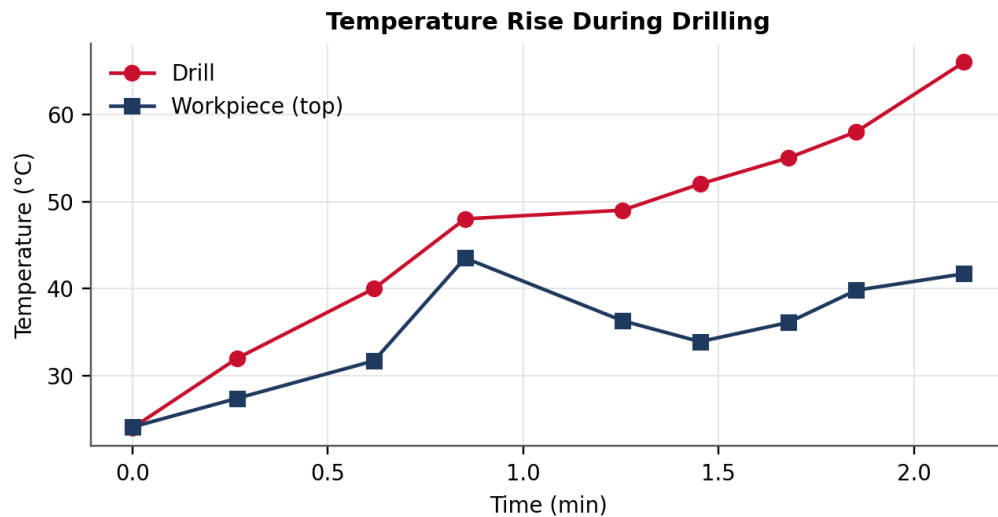


Figure 2. Drill and workpiece temperature over time.

5. Discussion

Feed rate, depth of cut, RPM, and the workpiece material all shape the final result. With the spindle speed held constant, adjusting only feed and depth of cut controlled the position and effect of each pass, which showed how machine settings can be tuned to a specific part requirement. The force data follows the expected trend: doubling the feed roughly tripled the cutting force at both depths, and the deeper 0.015 in cuts required about twice the force of the 0.007 in cuts at the same feed.

The temperature data confirmed that friction is the primary driver of heat buildup during cutting. The drill temperature climbed steadily from 24 °C to 66 °C in just over two minutes, while the workpiece peaked at 43.5 °C and fluctuated as the tool moved. This highlights the need for coolant or lubrication to prevent overheating, tool softening, and material damage. Minor discrepancies in the force calibration were noted, but the calibration was linear enough to support the comparisons above.

6. Conclusion

This project provided practical experience with the lathe, including turning, facing, and drilling. It covered the fundamentals of machining and the variables that must be controlled during production — heat, speed, and axis alignment in particular. Demonstrations under different settings showed how each parameter affects quality, and all project objectives were met.

7. Key Concepts & Calculations

Table 4. Worked machining calculations

Quantity	Relationship	Result
Spindle speed ($V = 90$ fpm, $D = 3.6$ in)	$N = 12V / (\pi D)$	96.2 RPM
Spindle speed ($V = 460$ fpm, $D = 3.293$ in)	$N = 12V / (\pi D)$	533.7 RPM
Cutting speed	$V = \pi DN / 12$	55 fpm
Material removal rate	$MRR = (\text{volume removed}) / \text{time}$	≈ 0.10 in ³ /min
Cutting time	$t = (L + A) / (f \cdot N)$	5 s
Cutting power	—	0.17 hp

How does tool temperature affect tool performance?

As tool temperature rises, tool hardness drops. A tool exposed to heat for long periods is progressively weakened and wears faster.

What is the difference between up milling and down milling?

In up (conventional) milling, the cutter rotation opposes the feed, so chip thickness starts at a minimum and grows to a maximum. In down (climb) milling, the rotation and feed move in the same direction, so the chip is thickest at the start of the cut and thinnest at the end.

How does heat generation affect power and tool wear?

Excessive heat raises power consumption and accelerates tool wear. Coolant and lubrication reduce this, and machine inspections and parameter calibration should be kept current.

What are the functions of the main lathe components?

The **lead screw** drives the tool longitudinally; the **back gears** allow low speeds and light feeds when high torque is needed; the **tailstock** supports drilling and other work on the rotational axis; and the **face plate** holds workpieces that cannot be mounted in a chuck.

How can a taper be turned on a lathe?

Common methods are offsetting (setting over) the tailstock, using a form tool, using the compound rest, and using a taper attachment.